

Evoltrya OS — Document 1: Requirements Discovery (Exhaustive Functional Breakdown)

Purpose

This is the first of three planning documents. Its objective is to lay out, as exhaustively as possible, the functional scope, open questions, inter-module relationships, and known challenges for every module of Evoltrya OS.

Once the open items below are resolved, this document feeds into **Document 2 (System Blueprint)** — covering architecture, module map, design principles, and current-state positioning — which in turn drives **Document 3 (Roadmap)** — a phased, executable build plan with priorities and milestones.

How to read this document

Every section flags items that require a business or domain decision. These use the following annotation conventions:

- **[CONFIRM]** — a working assumption that needs verification.
- **[DECISION]** — a rule, threshold, or process choice that must be made.
- **[SPECIFY]** — detail that requires business or industry input.
- **[DESIGN DECISION]** — an architectural fork with downstream consequences; decide with care.

Open items are intentionally left blank () for completion.

Overview: What Evoltrya OS Is

Evoltrya OS (formerly SWM-OS) is a **modular, extensible, permission-segregated enterprise operating system** for a lithium-battery recycling business spanning two legal entities in **Singapore and the EU**.

Core principles:

1. **Unified** — procurement, logistics, processing, inventory, sales, compliance, HR, finance, and business development are all integrated into a single OS.
2. **Extensible** — the number of modules is not fixed. New modules, whether core business functions or auxiliary ones (e.g. industry news, personal tasks), can be added continuously under a common standard.
3. **Permission-segregated** — each user can only view or edit the content their permissions allow.
4. **Multi-entity** — data is partitioned by Singapore/EU entity, with support for inter-entity operations.

5. **Functional depth first** — each module is to be built out fully, not left as simple transaction logging or data capture.

Current state (accurate assessment):

- **Framework foundations in place** — authentication, a bilingual (EN/ZH) i18n system, an atomic database transaction pattern, soft deletes, automatic code generation, state machines, and a validated module-development template.
- **Modules built (skeleton/ledger level only, not full-featured)** — suppliers, customers, materials (master data), plus inbound, output, processing, and inventory (a small part of the logistics and processing functions).
- **The majority is not yet built** — complete procurement, sales, finance, compliance/trade, HR, and BD modules, as well as the full business functionality of the modules listed above, all remain planned.

The system today amounts to a foundation, load-bearing structure, and a show unit. The show unit (suppliers/processing modules) demonstrates the standard for how a module should be built, but the dozens of actual rooms — full module functionality — remain shell or empty.

Part 1: Cross-Cutting Concerns (Foundations Underlying All Modules)

These are not individual modules but underlying mechanisms shared by every module. Their design constrains how each module is built, so they come first.

A. Permission System (RBAC)

Purpose: Controls who can do what, to which data, in which module. The basis of system-wide segregation.

Established direction: Permission implementation is deferred to the end, but the interface is reserved now — the data layer carries the "ownership" dimension and RLS hooks from the start, and the user-role table structure is established early. Once module functionality is built out, permission implementation is done in a unified pass.

Functional scope

Users and identity

- User account management (create, deactivate, password reset)
- User-to-entity association (a user belongs to SG, EU, or both)
- Login methods (email/password, SSO, multi-factor authentication)

[DECISION] Which login methods are required? Is MFA needed?

- ☐ Email + password (current)

- ☒ Google/Microsoft SSO
- ☐ MFA
- Notes: _____

Roles and permissions

- Role definitions (e.g. procurement officer, warehouse, processing supervisor, finance, sales, compliance officer, HR, administrator, owner/super-admin)
- Role-to-module permission matrix
- Operation-level permissions (view / create / edit / delete / approve / export, each independent)
- Field-level permissions (sensitive fields such as cost or margin visible only to certain roles)
- Data-scope permissions (own records only / own department / own entity / all)

[SPECIFY] What roles are envisaged, and roughly what can each do?

- Role 1: __ **access:** __
- Role 2: **access:** __
- ...

[DECISION] How granular should permissions be?

- ☒ Module level (can/cannot enter a module)
- ☐ Operation level (can enter but read-only)
- ☐ Field level (cost/margin hidden by role)
- ☐ Data-scope level (own entity/department/records only)

Data ownership (fields to reserve now)

- Each business record carries: owning entity (SG/EU), owning department/function, creator, responsible party
- Supabase RLS (row-level security) hooks: filter visible data by the user's entity/role

[DESIGN DECISION] How is the data "ownership" dimension defined? This determines which fields every table must carry.

- ☐ By entity (SG/EU) — almost certainly required
- ☒ By department/function (procurement, processing floor, ...)
- ☐ By project/business line
- Notes: _____

Approval workflows (interwoven with permissions)

- Configurable approval chains (different documents and amounts route to different approvers)
- Approval delegation (when an approver is away)
- Approval audit trail (who approved/rejected/countersigned, and when)

[SPECIFY] Which actions in the system require approval? (purchase orders, payments, price changes, stock issue, expense claims, ...) _____

All value related actions should have an approval

B. Multi-Entity Architecture (Singapore + EU)

Purpose: Two legal entities, with data both independent and able to interoperate. ^{deleted}

ease of compliance

Functional scope

- Entity master data (SG company, EU company: registration, tax IDs, bank accounts, functional currency)
- Data segregated by entity (by default only the user's entity is visible; authorized users may cross entities)
- Inter-entity operations:
 - Inter-entity material transfers (does material received in SG move to the EU for processing?)
 - Inter-entity internal transactions / transfer pricing (tax implications)
 - Consolidated reporting (financial roll-up across both entities)

[DESIGN DECISION] What is the business relationship between the two entities?

independent P&L entities can be pulled and compiled for group analysis

[SPECIFY] What does each entity do, and is there material/fund flow between them?

- SG entity primarily: *HA and non-OECD battery material*

- EU entity primarily: *OECD battery material*

- Between them: ☐ material transfers ☐ fund flow ☐ independent. Notes: _____

[DECISION] In which EU member state is the EU entity registered?

Poland, Italy, France for the start

- The EU has no single unified company registration; the entity is typically established in a member state (e.g. a German GmbH, Dutch BV, Polish entity, etc.), with tax regime, registration requirements, and local hazardous-waste regulations following that country.

- Registration country: _____ (affects tax-ID format, VAT rate, local compliance requirements)

[DECISION] Functional currency and multi-currency

- SG in SGD, EU in EUR? How are foreign-currency transactions (e.g. purchases/sales in USD) recorded and translated? *Biz related: all in USD; employee & admin: local currency*

- EU cross-border transactions also involve intra-EU VAT rules (VAT treatment between member states differs from China-EU / EU-third-country). _____

C. Audit and Traceability

Purpose: Full visibility into who changed what and when. Especially important given the recycling industry plus compliance requirements.

Functional scope

- Operation log (all create/update/delete: operator, time, before → after values)
- Document status-change history (e.g. the full progression of a processing run from draft to submitted)
- Soft deletes (in place, `deleted_at`, recoverable plus audit retention)
- Full batch traceability (a batch from inbound → processing → output → sale; basic chain in place)
- Audit-report export for key operations (for regulatory inspection)

[SPECIFY] What specific traceability requirements does compliance/regulation impose? (e.g. how many years must hazardous-waste records be retained?) _____

System architecture: only the system manager

Updates: functional leader (eg. pricing)

D. Coding System

Purpose: Unified numbering rules for documents and master data system-wide.

Current state: Automatic codes such as SUP - / CUS - / MAT - / IN - / OUT - / PROC - (generated by database triggers).

To extend

- Coding rules for new modules (PO, sales order, invoice, payment, transfer note, ...)
- Whether codes carry an entity identifier (e.g. SG-PO-2026-0001 vs EU-PO-2026-0001)
- Whether coding rules are configurable

[DECISION] Should codes reflect entity/year/module? Unified style: _____

Yes

E. Common Module Standard (Module Template)

Purpose: A unified structure followed by every module, ensuring consistency and maintainability. Existing suppliers/processing modules already follow it.

Standard per module:

- List page (filter, sort, search, paginate, export)
- Detail page
- Create/edit form (validation)
- Soft delete
- State machine (where applicable)
- Automatic coding
- Bilingual i18n
- Ownership fields (entity/department/creator)
- Operation log

[SPECIFY] What other common capabilities should every module carry as standard? (e.g. attachments, notes/comments, mentions, PDF/Excel export, bulk operations, data import, ...) _____

Part 2: Master Data

Existing suppliers/customers/materials fall here, but currently carry only basic fields; complete master-data management is considerably richer.

Module: Supplier Master Data

Current state: Basic fields + 8-state machine + compliance-certificate sub-table.

Scope to enrich

- Supplier classification (raw-material supplier / consumables supplier / service provider / logistics provider, ...) + *others to specify*
- Supplier qualifications and compliance (hazardous-waste handling licences, import/export permits, expiry alerts) — partly in place
- Supplier rating and evaluation (on-time delivery, quality pass rate, price competitiveness)
- Supplier contact management (multiple contacts, roles, details)
- Supplier bank accounts (for payment, multiple accounts)
- Supplier price file (historical quotes, agreement prices)
- Supplier transaction history (cumulative spend, running account)
- Blacklist/cooperation-status management (state machine in place)

[SPECIFY] What other supplier information must be managed? On what dimensions are suppliers evaluated? *good to go, and supplier audit process should be included here in a proper way*

[DECISION] (industry) For hazardous-waste intake, what hard requirements apply to supplier qualifications? Should the system block a purchase order when qualifications are expired? _____

Module: Customer Master Data

Current state: Basic fields (no state machine).

To enrich

- Customer classification (cathode maker, battery maker, trader, ...; can add credit rating, importance)
- Customer credit management (credit limit, terms, over-limit alerts)
- Customer contacts, shipping addresses (multiple)
- Customer bank/invoicing details
- Customer transaction history, accounts receivable
- Customer contract management

[SPECIFY] What else does customer management need? Does sales tier its customers? *good to go*

Module: Material Master Data

Current state: Category/chemistry/unit + CustomSelect free-text values.

To enrich

- Material coding (MAT- codes in place)
- Material specifications / technical parameters (chemical-composition ranges, form, purity grade)
- Material classification tree (multi-level, rather than a single-level category)
- Units of measure and conversions (kg/tonne conversion; units exist but no conversion relationships)
- Quality-inspection standards per material (what is tested for each)
- Material safety stock / reorder point
- Material relationships (e.g. which metals a given black mass yields)

[SPECIFY] (industry) What technical parameters must material master data record? What are the key attributes of black mass / metals / spent batteries? _____

[DECISION] Are unit-conversion relationships needed? (e.g. automatic 1 tonne = 1000 kg) Yes

Part 3: Core Business Modules

Module 1: Procurement

Purpose: The point of entry for material into the system. The full cycle from requirement to payment. Tightly coupled with logistics, finance, and inventory.

[CONFIRM] What is procured primarily? *1. Fixed assets (equipment, machines, etc.)*

- ☐ Recycled raw material (spent batteries / black mass) *2. Focused materials*
- ☐ Production consumables *- black mass* *3. Non-focused materials*
- ☐ Fixed assets / equipment *- waste batteries* *4. Peripherals*
- ☐ Other: _____ *- foils*
- repurposed batteries
- metals

Core flow

Purchase requisition (PR) → approval → purchase order (PO) → supplier confirmation → goods receipt (GR) → inspection → put-away → invoice matching → payment

[SPECIFY] What is the actual procurement action chain? → *They should be identical*

- Raw-material procurement flow: _____

- Consumables procurement flow: _____

- (Raw material may be "negotiate price → haul → weigh → pay", with only consumables running a full PR → PO?)

Functional scope

Purchase requisition (PR)

- Raise requisition (material, quantity, need date, purpose, budget)
- Multi-level approval (by amount/category)
- Status tracking (draft/pending/approved/converted to PO/rejected)
- Requisition consolidation (multiple PRs into one PO)

[DECISION] Approval levels and approvers? (e.g. < 10k by supervisor, > 10k by owner) *2 levels,*

Purchase order (PO)

- Generate PO from PR, or create directly
- PO lines: supplier, material, unit price, quantity, tax, delivery date, payment terms, Incoterm
- PO approval, send to supplier, confirmation receipt
- PO change management (quantity/price changes audited, versioned)
- Framework / blanket orders (agreed total, drawn in batches)
- PO linkage to supplier master data (carry over terms, historical prices)

*supervisor - approval
requisitor - raise request
≥ 10k : CFO approval*

Pricing and comparison

- Supplier quote entry, multi-party comparison
- Historical purchase-price traceability (price trend per material)
- Price-anomaly alerts (this price vs historical average)

[DESIGN DECISION] (industry) How is recycled raw material priced?

- ☒ By metal content / assay results (e.g. nickel-cobalt content × LME metal price)
- ☐ Fixed unit price
- ☐ Spot / negotiated per batch
- Notes: _____

- (Raw-material pricing pegged to metal markets is a capability standard ERP lacks and requires dedicated design.)

[DECISION] Integrate metal-market quotes (LME, SMM, etc.) as a pricing reference? *Yes*

Goods receipt (GR)

- Receipt against PO, partial receipt, over/short handling
- Weighbridge entry (by actual weighed weight; critical for raw material)
- Receipt linked to inbound batch

[DESIGN DECISION] Is goods receipt the same as the existing inbound batch?

- The current inbound module is standalone; in the full flow, inbound should be the result of a PO receipt.

- Decide: ☐ attach inbound under PO ☐ retain two intake paths (with PO / ad-hoc without PO)

- Direction: *There should be the ability to differentiate the discrepancies*

Incoming inspection

- Incoming inspection (assay composition, accept/concession/reject)

- Inspection results affecting pricing (price adjustment if actual content is below agreement)
- Two-stage pricing (receive at estimated price, adjust after assay)

[SPECIFY] (industry) What indicators are inspected on incoming material? Who tests, how long until results, how does it enter the system? _____

Reconciliation and payment (boundary with finance)

- Three-way matching (PO + GR + invoice)
- Discrepancy handling
- Payment requisition, payment plan, payment record
- Prepayment / deposit management

[SPECIFY] Payment terms for procurement? Prepayments / credit terms / instalments? *There should be a payment management system . eg.*

Procurement analytics

- Spend analysis (by material/supplier/period)
- Supplier performance (on-time, quality, price)
- Price trends

*deposit - 80%
arrival - 10%
post-weights - 10%*

Data relationships

→ logistics/inbound (receipt creates batch) / inventory (put-away) / finance (payable, payment) / supplier master data / compliance (import customs, hazardous-waste documents)

Known challenges

- Recycled material composition/weight uncertainty; pricing must adjust to actuals
- Two-stage pricing by assay results
- Compliance prerequisites for hazardous-waste procurement (verify qualifications, import permits)

[SPECIFY] What is the most painful/distinctive aspect of procurement? _____

- 1. chasing for manual approvals and signatures*
- 2. disparate data → all over the place and disconnected*

Module 2: Logistics

Purpose: The physical movement of material — transport, warehouse scheduling, stock in/out, cross-border logistics. The existing inbound/output is only the "batch record" part.

[CONFIRM] What does logistics scope include?

- ☒ Inbound logistics (supplier → warehouse)
- ☒ Outbound logistics (warehouse → customer)
- ☒ Plant / inter-warehouse transfers
- ☒ Cross-border logistics (SG ↔ EU, or international)
- ☒ Hazardous-waste transport (special qualifications)

Functional scope

Transport management

- Transport orders (pickup, destination, goods, vehicle/carrier, time)
- Carrier / freight-forwarder management (qualifications, quotes, performance)
- Transport mode (road/sea/air)
- In-transit tracking (status: pending/in transit/arrived/exception)
- Transport-cost accounting (freight, duties, insurance — capitalized into material cost?)

[SPECIFY] How is freight moved? Own fleet or outsourced? How is cross-border handled? _____

Warehouse management

local *international/local*

- Warehouse/bin master data (multiple warehouses, zones, bins)
- Inbound operations (receipt → put-away)
- Outbound operations (pick → ship)
- In-warehouse movement (transfer, repack)
- Hazardous-waste zone management (compliant-storage requirements)

[DESIGN DECISION] Manage down to bin level?

- ☐ Warehouse level only (which warehouse)
- ☒ Bin level (specific rack/zone)
- Notes: _____

[SPECIFY] (industry) What compliant-storage requirements for hazardous waste must the system reflect? *focused material & non-focused material*

Stock in/out documents

- Stock-in (purchase receipt, processing output, return, transfer-in)
- Stock-out (sales issue, processing draw, transfer-out, scrap)
- Stock-movement ledger

The existing inbound/output batches are essentially nascent stock-in/output records and need to be folded into a complete stock in/out system.

Packaging and labelling

- Batch label / barcode generation (weighbridge ticket, batch QR)
- Scan-based stock in/out

[DECISION] Should on-site work be scan-based? (weighbridge entry, scan-based stock in/out) *Yes if possible*

Data relationships

→ procurement (receipt) / processing (draw, output put-away) / sales (shipment) / inventory (movements update stock) / compliance (cross-border documents)

Known challenges

- Qualifications and documents for hazardous-waste transport
- Cross-border customs and tax
- Apportioning transport cost into batch cost

[SPECIFY] Most painful aspect of logistics? *1. discrepancies between actual cargo and online data*
2. mis-rep of nature of cargo

This module should be completely redesigned.

⇐ **Module 3: Processing / Production** ⇒ *The process you offered is more appropriate for building up a product, rather than reducing a product.*

Purpose: Converting inbound material into output. The current "input → output + loss" is only the most basic material balance; real production management is far richer.

Current state: Records processing runs (input legs / output legs), quantity conservation, loss, traceability chain, atomic commit/rollback.

Functional scope (well beyond current state)

Process and recipe

- Routing (which operations turn a material into output)
- Processing recipe / BOM (standard input-output ratios, standard loss rate)
- Multi-operation processing (crush → sort → leach → refine, ... each operation recorded separately?)

[DESIGN DECISION] (industry) Is processing single-step or multi-operation?

- Currently a single "input-output" step. In reality likely multi-operation (e.g. dismantle → crush → pyrolysis → hydrometallurgy, ...)
- ☐ Single-step (sufficient)
- ☐ Multi-operation (each operation records its own input/output/loss/labour)
- Actual process flow: _____

Production planning and execution

- Production plan / work order (what to process, how much, when)
- Work-order scheduling
- Actual-execution record (actual input/output vs plan)

Quality management *not needed*

- In-process inspection, finished-goods assay (metal content, purity, grade of output)
- Inspection results bound to output batch (purity field exists but unstructured)
- Non-conforming handling (rework/downgrade/scrap) ⇒ *not need.*

[SPECIFY] (industry) What indicators are tested on output? How is assay data recorded and linked to batches? _____

Cost roll-up

- Processing-cost accounting (raw material + labour + energy + equipment depreciation + consumables → apportioned to output batch)
- Standard cost vs actual cost

[DESIGN DECISION] Perform "output cost" accounting?

- This upgrades "logging" to "cost management". Involves apportioning electricity/labour/depreciation to each output batch.
- ☐ Yes (output must yield a unit cost, bearing on pricing and margin)

- ☐ Not yet
- Notes: _____

Equipment and labour

- Equipment register, run records
- Labour records (who spent how much time on this batch)

[SPECIFY] Manage equipment and labour? Are labour/equipment a large share of processing cost?

Yield and analysis

- Loss-rate analysis (loss in place; add trend/anomaly analysis)
- Yield analysis, metal recovery rate (input metal vs output metal — a core industry KPI)

[SPECIFY] (industry) What is the most-watched processing KPI? (recovery rate? loss rate? unit cost?)

Data relationships

→ logistics (draw, output put-away) / inventory (consume raw, add output) / procurement (raw-material cost) / finance (cost roll-up) / sales (output saleable)

Known challenges

- Batch traceability across multiple operations (recording intermediates)
- Composition conservation (not only weight but metal-element conservation)
- Complexity of cost apportionment

[SPECIFY] Most painful/distinctive aspect of processing management? _____

Module 4: Inventory / Stock

incoming
W.I.P
outgoing

Purpose: A real-time, accurate stock ledger. The current "remaining-quantity sum" is only the tip of the iceberg.

Current state: Raw/finished stock balance aggregated by material (read-only).

Functional scope

Stock ledger

- Real-time stock (by material/batch/bin/entity, multi-dimensional)
- Stock movement ledger (line detail of every in/out, each change traceable)
- In-transit stock, in-inspection stock, available stock (status-differentiated)

Batch and shelf life

- Batch management (batch concept in place)

- Batch shelf life / age (how long material has sat; hazardous waste may have storage time limits)
- FIFO / specified-batch issue

[SPECIFY] (industry) Do raw material/output have storage time limits? Should stock age be alerted?

Yes, we should be able to manage aging stock.

Stocktake

- Cycle count / spot count / full count
- Stocktake document, variance handling (adjustment and approval of book-vs-physical mismatch)
- Stocktake history

Quarterly or half-yearly random implied

[DECISION] Implement stocktake functionality? How often? *Yearly => Full stocktake*

Stock alerts

- Safety-stock alert (below reorder point prompts procurement)
- Overstock alert, slow-moving alert (long-idle material)

Stock valuation

- Stock value (weighted average / FIFO / moving average)
- Stock cost reconciled with finance

[DESIGN DECISION] Stock-valuation method?

- ☐ Weighted average
- ☐ FIFO
- ☒ Moving weighted average
- Notes: _____

- (Directly bears on the inventory value in financial statements; must align with accounting standards.)

Multi-dimensional analysis

- Inventory turnover
- Stock-structure analysis (by material/entity/age)

Data relationships

→ all stock in/out modules (procurement/logistics/processing/sales) update stock / finance (stock valuation posting)

Known challenges

- Accuracy of multi-status stock (available/in-inspection/in-transit/frozen)
- Precise batch-cost accounting
- Book-physical alignment (stocktake mechanism)

[SPECIFY] Most painful aspect of inventory management? *accuracy*

Module 5: Sales

Purpose: The full sales cycle for output. From inquiry to collection. Tightly coupled with inventory, logistics, finance, and customers.

Current state: Output batches have an optional customer field, but no sales process.

Core flow

Inquiry/quote → sales order (SO) → approval → shipment → invoicing → collection
before shipment

Functional scope

Quote management

- Customer inquiry, quotation
- Quotes based on cost / market price / metal quotes
- Quote history, quote-to-order conversion

[DECISION] (industry) How is output (metal/black mass) priced? Pegged to metal markets? *LME or SMM*

Sales order (SO)

- SO lines (customer, product, quantity, price, delivery date, payment terms)
- SO approval
- Inventory linkage (saleable-stock check, reservation)
- Contract management (long-term supply contracts, batched delivery)

Shipment

- Shipment notice, stock-out (integrating logistics/inventory)
- Partial shipment, shipment tracking

Invoicing and collection (boundary with finance)

- Sales invoice
- Accounts receivable, terms management
- Collection record, reconciliation
- Credit control (over-limit alert/freeze)

[SPECIFY] Is sales primarily spot or long-contract? How are terms managed? *Both, according to contract*

Sales analytics

- Sales performance (by customer/product/period/salesperson)
- Margin analysis (price – cost)
- Customer-contribution analysis

[SPECIFY] What does sales focus on most? (collection? margin? customer concentration?) *product focused*

Data relationships

→ inventory (saleable, issue) / logistics (shipment) / finance (receivable, collection) / customer master data / processing (output saleable) / compliance (export customs)

Known challenges

- Output pricing pegged to metal markets
- Margin accounting of sales vs cost (requires output cost first)
- Credit-risk control

[SPECIFY] Most painful aspect of sales? *Different pricing models should be integrated into this module.*

Module 6: Compliance & Trade

Purpose: The most distinctive module for the recycling industry. Covers hazardous-waste compliance, cross-border transfer, import/export customs, and regulatory filing.

[SPECIFY] (important, industry core) What does this module entail? Which specific compliance matters does it cover? How do frameworks such as the Basel Convention, the EU Battery Regulation, and IPCEI translate into system functionality? _____

Functional scope (general + industry, to be corrected)

Qualification and licence management

- Company qualifications (hazardous-waste handling licence, import/export rights, various permits, with expiry alerts)
- Supplier/customer qualification verification
- Material compliance classification (which are hazardous waste, and the corresponding control codes)

Cross-border transfer compliance

- Basel-related documents (prior informed consent for cross-border hazardous-waste transfer, etc.)
- Cross-border transfer application / approval management
- Full audit trail of transfer documents

[SPECIFY] What procedures apply to cross-border hazardous-waste movement (SG ↔ EU or international)? Which documents must the system manage? _____

Import/export customs

- Customs declaration management (import/export)
- HS codes, duties, rebates
- Freight-forwarder / customs-broker integration
- Document management (B/L, invoice, packing list, certificate of origin, ...)

Regulatory filing

- Regulatory filing per jurisdiction (periodic/per-transaction filings to environmental authorities and customs)
- Filing data auto-generated from business data (traceability chain → filing forms)
- Filing records and acknowledgements

[SPECIFY] To which agencies, at what frequency, must what be filed? (the core of the compliance module; requires domain expertise) _____

Regulation library and alerts

- Applicable-regulation list (Basel, EU Battery Regulation, SG/EU local hazardous-waste regulations, ...)
- Regulation-update reminders
- Compliance checklist (per-transaction self-check)

[SPECIFY] Which regulations directly govern the business? Which must be observed day to day?

Data relationships

→ procurement (import-material compliance) / sales (export-product compliance) / logistics (cross-border transport) / suppliers and customers (qualifications) / traceability (filing-data source)

Known challenges

- Highly specialised; complex, multi-jurisdiction regulation
- Filing-data accuracy and traceability (directly facing regulatory inspection)
- Fast-changing regulation; the system must adapt

[SPECIFY] Most painful/distinctive aspect of compliance? _____

Module 7: Finance ⇒ see *"Limitations of Xero for a Battery Recycling Business"*

Purpose: Accounting functionality on par with Xero, while solving the pain points Xero cannot. This is the key to upgrading the OS from a business system to a management system.

[SPECIFY] (core) What specifically are the pain points Xero cannot solve? This is the design soul of the finance module. List in as much detail as possible:

- Pain point 1: ____

- Pain point 2: ____

- Pain point 3: ____

- (e.g. Xero does not understand batch cost? does not support metal-based pricing for recycling? multi-entity consolidation is cumbersome? disconnected from business data, requiring duplicate entry?)

Functional scope (general accounting + pain-point resolution)

General ledger (GL)

- Chart of accounts (two sets for SG/EU?)
- Journals (manual journals + auto-generated from business)
- Period management (month-end, year-end, period lock)
- Multiple ledgers / multiple entities

[DESIGN DECISION] Depth of finance-to-business integration?

- Xero's pain points often lie in business and finance being disconnected, requiring duplicate entry. Evoltrya OS's advantage is precisely that business data is native to the system.

- ☐ Business documents auto-generate journals (purchase → payable journal, sales → receivable journal, stock movements → inventory journal, ...)
- ☐ Finance keeps its own books, weakly linked to business
- Expectation: _____

Accounts receivable / payable (AR/AP)

- Accounts payable (purchase invoices, payment clearing, ageing)
- Accounts receivable (sales invoices, collection clearing, ageing, chasing)
- Statement reconciliation (supplier/customer statements)

Treasury

- Bank accounts, bank statements (import/reconcile)
- Cash flow (actual + forecast)
- Payment plans and approvals

Cost accounting (deeply coupled with processing/inventory; a Xero weakness)

- Batch cost, product cost (raw material + processing + apportionment)
- Stock valuation and cost rollover
- Margin analysis (price – cost)

[SPECIFY] Is cost accounting a primary motivation for building this finance module? What numbers do you want to get straight? _____

Tax

- SG GST / EU VAT tax handling
- Tax reports
- Multi-currency and exchange rates

[DECISION] Are the SG and EU tax regimes handled separately? Are foreign currencies (USD transactions) involved? _____

Reporting

- The three statements (balance sheet, P&L, cash flow)
- Multi-entity consolidated statements
- Management reports (custom operating analysis)
- Budget vs actual

[SPECIFY] What reports/insights do you most want from finance that Xero cannot provide? _____

Relationship with external tools

- Does it coexist/sync with Xero, or fully replace it?
- Report export (for accountant/audit/tax)

[DESIGN DECISION] Relationship of the Evoltrya OS finance module to Xero?

- ☐ Fully replace Xero
- ☐ Partial replacement; keep core bookkeeping in Xero with data sync
- ☐ Evoltrya does business + cost, Xero does statutory books, the two integrate
- Direction: _____

Data relationships

→ nearly every module (procurement → AP, sales → AR, processing/inventory → cost, HR → payroll, logistics → expense) / multi-entity consolidation

Known challenges

- Accounting-standard compliance (SG/EU differ)
- Journal automation between business and finance (both an advantage and a complexity)
- Precision of cost accounting
- Trade-offs of replacing/integrating Xero

[SPECIFY] Most painful/most-desired-to-solve aspect of finance? (please be detailed; this is a key module) _____

Module 8: HR *⇒ must be compliant to local requirements, for now, it's Singapore*

Purpose: HR execution is outsourced, but data is integrated within the system. The outsourcer handles transactions, while employee data, attendance, payroll, etc. are retained in Evoltrya OS.

[CONFIRM] What does "outsourced but data-integrated" specifically mean?

- ☐ The outsourcer operates within Evoltrya OS (granted access)
- ☐ The outsourcer operates in their system, syncing data into Evoltrya OS
- ☐ Evoltrya OS stores only key HR data for internal view, with transactions external
- Notes: _____

Functional scope *⇒ End results will be provided as raw data*

Employee master data

- Employee records (basic info, position, on/off-boarding, contract, documents)
- Organization structure (departments, reporting lines — also supports the permission system)
- Multi-entity employees (separate SG/EU workforces)

Attendance and scheduling

- Attendance records, scheduling (shift work on the processing floor?)
- Leave, overtime

Compensation

- Payroll calculation (base + overtime + performance + deductions)
- Payroll-disbursement records
- Integration with finance (payroll posted as cost/expense)

[SPECIFY] Is payroll calculated by the outsourcer or the system? Does the processing floor have piece-rate / shift pay? _____

Performance and other

- Performance appraisal
- Training, certificates (hazardous-waste operation may require certification)

[SPECIFY] (industry) Must processing / hazardous-waste staff hold certificates? Must certificate validity be managed? _____

Data relationships

→ permission system (org structure, positions) / finance (payroll cost) / processing (labour) / compliance (certified to operate)

[SPECIFY] What does the HR module most need? Given outsourcing, what is most important to retain internally? _____

Module 9: Business Development (BD)

Purpose: Business development — new supplier/customer development, cooperation projects, opportunity management. Between sales and strategy.

[CONFIRM] What does BD specifically do here?

- ☐ Develop new raw-material supply channels
- ☐ Develop new customers / new markets
- ☐ Strategic cooperation / joint ventures
- ☐ Other: _____

Functional scope (general CRM/BD view)

Opportunity / lead management

- Lead entry and source
- Opportunity stages (contact → negotiation → intent → close)
- Follow-up records, activity log

Cooperation-project management

- Project register (partner, stage, key milestones)
- Project documents, contracts

Contact / relationship management

- Prospective supplier/customer pool
- Contacts, interaction history

Conversion and handoff

- BD outcomes converted to formal supplier/customer master data
- BD → procurement/sales handoff

[SPECIFY] What is the BD workflow? What is managed and what is it converted into? _____

[SPECIFY] What does BD most need from the system? _____

Data relationships

→ supplier/customer master data (conversion) / procurement/sales (realization) / document management (contracts)

Part 4: Auxiliary / Common Modules (Open-Ended, Not Limited to These)

These are not core business functions but improve overall experience and efficiency. The design must allow this class of module to grow continuously.

Module: Dashboard / Workbench

- Personal workbench (my tasks, my documents, pending my approval)
- Operating dashboard (stock levels, procurement/sales/processing overview, finance overview)
- Role-tailored home page (procurement sees procurement, owner sees the whole)
- KPI visualization (recovery rate, loss, turnover, margin, ...)

[SPECIFY] What do you and each role most want to see on the home page? _____

Module: Tasks / To-Do

- Personal tasks (manual + system-derived: pending approval, pending receipt, pending payment, ...)
- Task assignment and collaboration
- Reminders and due-date alerts
- Linkage with business modules (e.g. "this PO awaits your approval" auto-becomes a task)

[SPECIFY] Are tasks mainly personal, or team task collaboration? _____

Module: Industry News / Daily Updates

- Industry-news aggregation (metal quotes, policy and regulation, market dynamics)
- Daily push / digest
- Linkage with business (e.g. metal prices fed directly to procurement/sales pricing?)

[SPECIFY] Where does news come from? What is aggregated? (LME metal prices? policy?) Should it feed the pricing module? _____

Module: Notification Center

- System notifications (approvals, alerts, expiries, exceptions)
- Multi-channel (in-app, email, instant messaging such as Slack/WhatsApp?)
- Notification preferences

[DECISION] Where should notifications be pushed? (in-app/email/WeChat/WhatsApp) _____

Module: Document Management

- Document attachments (contracts, customs declarations, assay reports, certificates, ...)
- Document classification, search
- Version management, access control

[SPECIFY] Which documents most need systematic management? _____

Module: Reporting Center

- Cross-module reports
- Custom reports / queries
- Scheduled reports, export (Excel/PDF)

[SPECIFY] Beyond each module's built-in reports, is a unified reporting center needed? Which reports are most used? _____

[SPECIFY] What other modules/features are wanted that are not mentioned above? (The system is open and extensible — list freely.) _____

Part 5: System-Wide Questions (Requiring Holistic Decisions)

These do not belong to a single module but affect how the whole system is built.

1. Module rollout priority

[DECISION] Across the core and auxiliary modules, what is the priority order? Which must be usable first? *1st. manufacturing 2nd. warehousing 3rd. Finance 4th. Procurement + Sales*

2. Enriching existing modules vs new modules

[DECISION] Build the existing logistics/processing/inventory from "skeleton" to "full functionality" first, or roll out new modules (e.g. finance/sales) first? _____

3. Data migration

[SPECIFY] Is there historical data (Excel/Xero/legacy system)? Should it be migrated into Evoltrya OS?

No

4. Rollout cadence and parallel use

[DECISION] Build-and-use incrementally, or build a batch then cut over? How do existing Excel/Xero processes transition? Maximum: 50 users

5. Performance and scale

[SPECIFY] Expected data volume? (How many inbound/processing/sales transactions per month? How many users?) _____

- (Affects whether to remain on the Supabase free tier, and architecture choices.)

6. Mobile

[DECISION] Which scenarios require mobile use? (on-site weighing, scanning, approval, viewing reports?) Shopfloor activities first.

7. External-system integration

[SPECIFY] Beyond Xero, what else needs integration? (banks, customs, metal markets, logistics-provider systems?) LMZ, Searates, etc.

8. Budget and timeline

[SPECIFY] Is this built incrementally with AI assistance, or are there time/resource constraints? What milestone is expected by when? _____

Appendix: Using This Document

1. **Complete each section** — fill in every flagged item ([CONFIRM] / [DECISION] / [SPECIFY] / [DESIGN DECISION]) with real business, industry, and decision information.
2. **Correct assumptions** — change anything that does not match reality.
3. **Trim what is not needed** — mark unused general functionality as "not required".
4. **Add what is missing** — add anything needed but not anticipated.

Once completed, this feeds into **Document 2 (System Blueprint)** — architecture, module map, design principles, precise current-state positioning — and then into **Document 3 (Roadmap)** — a phased, executable build plan with priorities and milestones.

This document's value is in decomposing the large goal of building a complete enterprise OS into the concrete dimensions of each module, and clearly flagging the items that require business and domain input. It is a starting point for shared understanding and collaborative design, not a final answer.

Limitations of Xero for a Battery Recycling Business

1. Xero is accounting-driven, not operations-driven

Xero is primarily designed to record financial transactions such as invoices, bills, payments, bank reconciliations, and financial reports. It works well as a general accounting system, but it is not designed to manage the operational reality of a battery recycling business.

In battery recycling, financial results are generated from physical material movements, production conversion, batch yields, compliance requirements, and waste-handling decisions. Xero can record the final accounting outcome, but it does not naturally explain how that outcome was created through each battery, pallet, batch, process, and permit.

2. Limited support for raw materials and work-in-progress

Battery recycling is not a simple buy-and-sell business. A company may receive used batteries, sort them, discharge them, dismantle them, crush them, dry them, and separate them into black mass, copper, aluminium, plastic, and residue.

This process requires raw material tracking, work-in-progress tracking, production orders, yield calculation, loss calculation, and output batch creation. Xero's native inventory function is not designed to manage raw materials and work-in-progress in this level of detail.

As a result, Xero may record inventory value and cost of goods sold, but it cannot naturally manage the full production transformation from input batteries to multiple output materials.

3. Weak batch, pallet, and container-level traceability

Battery recycling requires strong traceability. A company needs to know where each batch came from, which supplier provided it, which container it arrived in, which pallet it was stored on, whether it was soaked or non-soaked, whether it had voltage, whether it was suitable for repurposing or recycling, which production line processed it, and which output batch was produced from it.

Xero is not designed for this type of detailed material lifecycle tracking. It can store invoices, bills, attachments, and basic item information, but it does not naturally provide a full traceability chain such as:

Supplier Batch

→ Container

→ GRN

→ Pallet

→ SKU / Material Category

→ Work Order

→ Output Batch

→ Sales Invoice

→ General Ledger

For a battery recycling business, this traceability chain is critical for internal control, compliance, production analysis, and external audit.

4. Difficulty in handling one-to-many production conversion

In battery recycling, one input batch can generate several different output materials. For example, one batch of used lithium-ion batteries may produce black mass, copper, aluminium, plastics, and waste residue.

Xero does not naturally handle this one-to-many production conversion. It is not designed to calculate production yield, allocate input cost across multiple output materials, or manage by-products and production losses in an industry-specific way.

This makes it difficult to answer key operational questions such as:

- How much black mass was produced from this battery batch?
- What was the yield percentage?
- How much copper and aluminium were recovered?
- What was the production loss?
- Which output product should carry which portion of the input cost?

5. Limited ability to calculate accurate batch-level COGS

The cost of goods sold in battery recycling is more complex than ordinary trading. It may include:

- Battery purchase cost
- Freight and shipping cost
- Local transportation
- Unloading and handling
- Sorting labour
- Discharging cost
- Electricity consumption
- Gas, nitrogen, or vacuum system cost
- Crushing and drying cost
- Production labour
- Maintenance consumables
- Waste treatment cost
- Rental and overhead allocation
- Losses during production
- Offset value from by-products

Xero can record these expenses at the accounting level, but it does not automatically allocate them by GRN, work order, material type, production batch, output product, yield, weight, or metal content.

This creates a major limitation: the company may know its overall profit and loss, but it may not know the true gross margin of each batch of black mass or each production run.

6. Not designed for metal-content-based inventory valuation

In battery recycling, the value of inventory is not determined only by weight or purchase price. It is also affected by metal content, moisture, impurity level, chemistry type, recoverability, and market metal prices. For example, two batches of black mass may both weigh 10 tonnes, but their values may be very different if one has higher cobalt and nickel content than the other.

Xero does not naturally link inventory valuation to assay results, metal percentage, moisture level, impurity level, or live market price assumptions. Therefore, the inventory value recorded in Xero may be acceptable for basic accounting, but it may not reflect the true commercial value of the material.

7. Limited support for repurpose vs recycle classification

Battery recycling companies often need to classify incoming batteries into different treatment paths. Some batteries may be suitable for repurposing, while others must be recycled. This classification may depend on voltage, internal resistance, physical condition, water exposure, testing records, and customer requirements.

Xero does not provide a native structure to manage repurpose vs recycle decisions at batch or pallet level. These details may be stored manually in descriptions or attachments, but they are not part of an integrated operational workflow.

This creates a gap between the accounting record and the technical decision-making process.

8. Difficulty in managing disposal-fee business models

Battery recycling businesses may operate under different commercial models:

- The company pays suppliers to purchase battery materials.
- The company receives batteries for free.
- The customer pays the company a disposal or treatment fee.
- The same batch may generate both service income and material resale value.

Xero can record invoices and bills for these transactions, but it does not naturally manage the mixed commercial logic behind them. A single batch may need to carry disposal income, inventory value, processing cost, and later sales revenue.

Without an industry-specific ERP layer, these transactions may require manual judgement and manual journal entries, increasing the risk of inconsistency.

9. Compliance data is not deeply integrated

Battery recycling involves compliance information that ordinary accounting systems do not manage in detail, such as:

- HS code
- Basel classification
- Import and export permits
- NEA or DOE approvals
- UN number
- MSDS
- Packing declaration
- Non-hazardous declaration
- Country of origin
- Country of destination
- Transit route

- Container number
- Seal number
- Soaked or non-soaked status
- Voltage status
- Internal resistance status
- Repurpose or recycle classification

Xero can store documents as attachments, but it does not understand these compliance fields as structured operational data. It cannot easily validate whether a shipment, batch, or sales transaction has the required compliance documents before proceeding.

For a battery recycling company, this is a significant limitation because compliance is not just documentation; it is part of the operating workflow.

10. Limited connection between physical inventory and financial records

External auditors do not only need invoices and general ledger reports. For a battery recycling business, they may also need to understand how physical stock movements support the accounting numbers.

They may ask:

- Why did 100 tonnes of input batteries produce only a certain amount of black mass?
- Where are the remaining copper, aluminium, plastics, or residues?
- Were the production losses reasonable?
- Does the stock count match the system records?
- Does the sales quantity match the output batch?
- Does the inventory valuation match the production records?

Xero can provide accounting reports, but it does not naturally provide the complete chain from GRN to pallet, work order, output batch, goods issue note, sales invoice, and general ledger.

11. Weak support for operational audit trail

Xero has accounting audit trail functions, but a battery recycling business needs more than accounting audit trail. It needs operational audit trail.

For example, the system should record:

- Who received the material
- Who weighed it
- Who classified it
- Who changed the material category
- Who approved the work order
- Who recorded the output weight
- Who approved the production loss
- Who released the batch for sale
- Who uploaded the compliance documents
- Who changed the valuation assumption

Xero is not designed to track all these operational actions in a single material lifecycle. This limits its usefulness as a full audit foundation for a recycling business.

12. Limited production performance analysis

Battery recycling management requires production analysis, not just accounting reports.

Management needs to know:

- Yield by battery type
- Yield by supplier
- Yield by production line
- Black mass output by batch
- Copper and aluminium recovery rate
- Processing cost per tonne
- Electricity cost per tonne
- Labour cost per tonne
- Gross margin by work order
- Gross margin by supplier
- Gross margin by product type

Xero is not designed to generate these industry-specific operational and production KPIs. These reports usually require a separate ERP, MES, WMS, or custom database.

13. Data fragmentation when using third-party add-ons

One common way to extend Xero is to connect it with third-party inventory, manufacturing, warehouse, HR, payroll, document, or reporting applications.

However, this can create data fragmentation. Different systems may hold different versions of inventory quantity, production status, cost, compliance documents, and financial records.

For battery recycling, fragmented data is risky because one transaction may affect inventory, production, compliance, GST, sales, cost allocation, and audit evidence at the same time.

A unified ERP can reduce this risk by keeping all operational and financial data in one database.

14. Limited support for multi-country and multi-entity recycling operations

Battery recycling often involves cross-border material movement, overseas suppliers, overseas customers, transit shipments, bonded or non-bonded arrangements, different permits, and different operating entities.

Xero can support multi-currency and separate entities, but it is not designed to manage cross-border recycling workflows such as:

- Shipment ownership transfer
- Intercompany inventory movement
- Transit cargo status
- Permit status by country
- Factory processing in another jurisdiction
- Cross-border cost allocation
- Entity-level and group-level inventory visibility

This makes Xero less suitable as the central operating system for a regional battery recycling group.

15. Limited ability to support management decision-making

Xero is useful for financial reporting, but battery recycling management requires decision-making based on physical and technical data.

Management may need to decide:

- Which supplier provides the best yield?
- Which battery chemistry is most profitable?
- Which production line has the lowest processing cost?
- Which batch should be processed first?
- Which material should be sold immediately?
- Which black mass batch should be held because of metal price movement?
- Which customer or supplier creates the highest compliance risk?

Xero does not naturally combine financial, operational, technical, and compliance data to support these decisions.

Conclusion

Xero can record the accounting results of a battery recycling business, but it is not designed to manage the full operational process that creates those results.

For a battery recycling company, the key limitation of Xero is that it starts from accounting documents, while the business reality starts from physical materials.

A purpose-built ERP for battery recycling should be designed around:

- Battery batch traceability
- Container and pallet control
- GRN and goods issue records
- Work orders and production conversion
- Yield and loss calculation
- Black mass, copper, aluminium, and residue tracking
- Batch-level COGS
- Metal-content-based valuation
- Compliance document control
- Repurpose vs recycle classification
- Audit trail and period lock
- Integration between inventory, production, sales, finance, and audit reports

In simple terms:

Xero records financial outcomes.

A battery recycling ERP should explain where every dollar came from, through every battery, batch, pallet, process, permit, and production result.